

Logic note

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1 Introduction

Most of the information i will introduce is from books Martin Hils', Jeremy Avigad's.

2 Language and logic

처음에 영어 책을 따라쓰다보니 영어로 적게 되었네요.. 해당 노트는 논리학 입문을 위해 작성했습니다. 사실 2.4부터 읽으시는걸 추천드립니다. 논리는 퍼즐이 아닙니다. 인간의 직관으로 세운 학문이기에 충분히 이해할 수 있고 직관적으로 접근할 수 있는 분야입니다. 해당 노트에서는 그게 어떻게 가능한지 천천히 설명해보고자 합니다. 기호가 굉장히 많지만 읽다보면 점점 익숙해지고 직관적으로 이해가 되실 겁니다.

논리체계에는 두가지 종류가 있습니다. 첫번째는 Propositional logic으로, 명제의 참거짓만 말하고, 그 명제에서 어떤 원소가 존재하는지, 성립시키는지에 대해서는 이야기 하지 않습니다. 고등학교 집합과 명제에서는 이 Propositional logic을 다룹니다.

두번째로 배우는 First order logic은 대학 수학에서 처음 배웁니다. 미적분학 시간에 For all, there exists 같은 말들은 모두 first order logic에서 왔습니다. 이 논리체계는 명제의 참거짓 뿐만아니라 그 명제를 만족시키는 원소, 원소끼리의 같음, 원소끼리의 관계에 대해 다룹니다. 더 나아가 Second order logic도 할 수가 있고, 원소들의 집합에 대해서도 이야기 합니다.

본 노트에서 초반 부분은 논리학 교양시간에 배우는 내용과 겹치는데요, 거기서 더 나아가 조금더 어렵지만 흥미로운 내용을 다루려고 합니다. 특히 모든 참인 명제는 증명 가능한가? 또는 특정 논리 체계를 약화시킬때는 얼마나 증명가능한가?에 대해서 직접 증명을 해보고자 합니다. 그리고 수학을 엄밀하게 하고 싶으신 분들을 위해, 다양한 역설을 소개하고 그 역설을 어떻게 해결할 수 있는지 배워보고자 합니다.

이 노트의 최종 목표는 집합론을 논리의 시각에서 이해하는 것, 조금더 자세히 하면 러셀의 역설이나 칸토어의 역설을 다루면서 엄밀한 논리를 도입하여 어떻게 모순과 역설을 벗어나는지 배워보고 무한수에 대해 논리적인 시야로 이해를 해보는 것입니다.

2.1 Language of logic

Why is language needed? we will make several rules and proofs in a logic system, and we need to 'use' some language so we should be careful about it.

Definition 2.1. A *language* is a 4-tuple $(\Gamma, \Delta, a, \mathcal{V}, \mathcal{C})$ where Γ and Δ are disjoint sets of symbols and a is function from $\Gamma \cup \Delta$ to $\mathbb{N}$. Γ is said to be set of *function symbols*, Δ is that of *relation symbols*.

a assigns *arity* of function and relation symbols, loosely speaking it is defining the 'input range' of symbols. For example 2-ary function symbol f takes two symbols $f(x_1, x_2)$.

and $\mathcal{V}$ is set of variable symbols, it can be countable or uncountable. and for function symbol with arity 0, it can be read as constant symbol.

2.2 Propositional logic

Propositional logic is maybe the first deductive logic system people ever made. It is known that Chrysippus made deduction system for propositional logic, in the BC.3 . Aristotel made syllogism logic system, which is later replaced by first order logic, or predicate logic by Gottlob Frege, particularly in his book Begriffsschrift. First introduction of symbol on propositional logic was 17 or 18th century, it was made by Leibniz, and developed by George Boole and De Morgan, independent of work of Leibniz's. Frege and Russell after influenced inventing truth table.

There are symbols in propositional logic, namely propositional variable and connectives, and $\perp$. Now we can define how to construct 'formula' from those variables and connectives.

Definition 2.2. Formula of propositional logic

- If P is propositional variable, then P is a propositional formula.
- $\perp$ is a propositional formula.
- If A and B are propositional formula, so are $(A \wedge B)$, $(A \vee B)$, and $(A \rightarrow B)$.

If there are no confusion, we can remove parentheses. Now we can define Complexity, Depth, variables and subformulas function inductively:

Definition 2.3.

- $\text{complexity}(\perp) = \text{complexity}(P) = 1$
- $\text{complexity}(A \circ B) = \text{complexity}(A) + \text{complexity}(B) + 1$
- $\text{depth}(\perp) = \text{depth}(P) = 0$
- $\text{depth}(A \circ B) = \max(\text{depth}(A), \text{complexity}(B)) + 1$
- $\text{variables}(\perp) = \emptyset$
- $\text{variables}(P) = \{P\}$
- $\text{variables}(C \circ D) = \text{variables}(C) \cup \text{variables}(D)$
- $\text{subformulas}(\perp) = \perp$
- $\text{subformulas}(P) = \{P\}$
- $\text{variables}(C \circ D) = \text{subformulas}(C) \cup \text{subformulas}(D) \cup \{C \circ D\}$

2.3 First order logic

First order logic(Predicate logic) is foundational logic system for mathematics. It has additional to propositional logic, variable and constant. For that, we can make more meaningful formula such as $P(x)$. I will talk about its usefulness over propositional logic in later chapter. We can talk about set theory of first order logic(Not exactly, we can't actually define 'set' in this first order predicate logic). To define formula in first order logic, we first need to define *terms* inductively.

Definition 2.4. Inductive definition of term

- Each variable x is a term
- If $c \in \Gamma$ is a constant symbol, then it is a term.
- If $f \in \Gamma$ is a k -ary function, then if $t_0, \dots, t_{k-1}$ are terms, then so is $f(t_0, \dots, t_{k-1})$.

Now we give definition of formula in first order logic.

Definition 2.5. Inductive definition of term

- $\perp$ is a formula.
- If s and t are terms, then $s = t$ is a formula.
- If $t_0, \dots, t_{k-1}$ are terms, then for k -ary relation symbol $R \in L$, $R(t_0, \dots, t_{k-1})$ is formula.
- If A and B are formulas, then so are $(A \wedge B)$, $(A \vee B)$, and $A \rightarrow B$.
- If A is a formula, x is a variable, then $\exists xA$ or $\forall xA$ are formula.

2.4 Deduction rules of propositional logic

There are several different rule system. First we introduce rules for propositional calculus, Hilbert's axiom. And then we introduce another deduction system, by Gentzen. Here we briefly mention Hilbert's one.

1. $A \rightarrow (B \rightarrow A)$
2. $(A \rightarrow (B \rightarrow C)) \rightarrow ((A \rightarrow B) \rightarrow (A \rightarrow C))$
3. $A \rightarrow (B \rightarrow A \wedge B)$

4. $A \wedge B \rightarrow A$
5. $A \wedge B \rightarrow B$
6. $A \rightarrow A \vee B$
7. $B \rightarrow A \vee B$
8. $(A \rightarrow C) \rightarrow ((B \rightarrow C) \rightarrow (A \vee B \rightarrow C))$

This is Hilbert's system. and there are three different logical systems, namely minimal, intuitionistic, classical logic. What's the difference? They have different logic axioms. For minimal, we add **Modus ponens**, which is rule From $A \rightarrow B$ and A , we can conclude B . For the intuitionistic, we add axiom

9. $\perp \rightarrow A$.

This is known as *ex falso sequitur quodlibet*, which means "from falsity, anything follows." For classical logic, we replace axiom 9 with following axiom:

- 9' $\neg\neg A \rightarrow A$.

But what does it mean for $\neg\neg A \rightarrow A$? In intuitionistic logic, it can be interpreted as 'if we can find contradiction from one's nonexistence'.

There are lots of examples using this tactic. Most famous one will be proof of non-rationality of square root of 2.

And if we were to take $\top, \neg, \leftrightarrow$ as basic connectives as well, we add following axioms:

$$\begin{aligned}
& : \top \\
& : (A \rightarrow \perp) \rightarrow \neg A \\
& : \neg A \rightarrow (A \rightarrow \perp) \\
& : (A \leftrightarrow B) \rightarrow (A \rightarrow B) \\
& : (A \leftrightarrow B) \rightarrow (B \rightarrow A) \\
& : (A \rightarrow B) \rightarrow ((B \rightarrow A) \rightarrow (A \leftrightarrow B))
\end{aligned}$$

Now we introduce natural deduction of Gentzen, which contains more rules than that of Hilbert's. and it is named natural, because it feels natural to use these rules when we do logic. Rules are quite intuitive. There are two kinds of rules, introduction rules and elimination rules. And those lines of tree-like proofs are called sequent. Try to memorize all the rules.

$$\begin{array}{c}
\frac{A \quad B}{A \wedge B} \wedge I \quad \frac{A \wedge B}{A} \wedge E_l \quad \frac{A \wedge B}{B} \wedge E_r \quad \frac{\overline{A} \quad \vdots \quad B}{A \rightarrow B} \rightarrow I \\
\frac{A \rightarrow B \quad A}{B} \rightarrow E \\
\frac{A}{A \vee B} \vee I_l \quad \frac{B}{A \vee B} \vee I_r \quad \frac{\overline{A} \quad \overline{B} \quad \vdots \quad C \quad \vdots \quad C}{A \vee B \quad C} \vee E
\end{array}$$

All rules above are sound for minimal logic, in other words, it is compatible with Hilbert axiom and modus ponens. For intuitionistic, add following:

$$\frac{\perp}{A} \perp E$$

This can be viewed as elimination rule for $\perp$. For classical logic, we add axiom(*reductio ad absurdum*):

$$\frac{\overline{\neg A} \quad \vdots \quad \perp}{A}$$

And as $\neg A$ is abbreviation for $A \rightarrow \perp$, we have following:

$$\frac{\frac{\overline{A}}{\vdots} \frac{\perp}{\neg A} \neg I}{\frac{\neg A}{\perp} \neg E} \neg E$$

Similarly, we have following for True $\top$, we have following axiom:

$$\begin{array}{c} \overline{A} \quad \overline{B} \\ \vdots \quad \vdots \\ \frac{B \quad A}{A \leftrightarrow B} \leftrightarrow I \end{array} \quad \frac{A \leftrightarrow B \quad A}{B} \leftrightarrow E_l \quad \frac{A \leftrightarrow B \quad B}{A} \leftrightarrow E_r$$

As you can see, minimal logic can't prove some formulas that classical or intuitionistic logic can. And only classical logic can do proof by contradiction.

Now, we can define what the 'proof' is.

Definition 2.6. A proof with hypothesis Γ is a sequence $(B_0, \dots, B_{n-1})$ of formulas such that for each $k < n$, either B_k is a formula in Γ , or B_k is an instance of an axiom schema, or there are $i, j < k$ such that B_k follows from B_i and B_j by modus ponens. If d is a proof with hypothesis Γ and A is the last element of d , then A is said to be the conclusion of d , denoted by $\Gamma \vdash A$.

Here, we didn't specify the type which formula it is supposed to be in, since it can both defined on propositional and predicate logic, respectively.

2.5 propositional inference rule(Fitch-style proof)

- **Simplification (simp)** From $A \wedge B$, we can deduce A and B .

$$A \wedge B \vdash A, A \wedge B \vdash B$$

- **Conjunction (conj)** From A and B , we can deduce $A \wedge B$.

$$\{A, B\} \vdash A \wedge B$$

- **Addition (Add)** From A , we can deduce $A \vee B$ and $B \vee A$ for any formula B .

$$A \vdash A \vee B, A \vdash B \vee A$$

- **Modus ponens (MP)** From $A \rightarrow B$ and A , we can deduce B

$$\{A \rightarrow B, A\} \vdash B$$

- **Conditional proof (CP)** If we can deduce B from assumption A , we can deduce $A \rightarrow B$.

$$A \dots B \vdash A \rightarrow B$$

- **Constructive Dillema (CD)** If we have $A \vee B$, and we have $A \rightarrow C$ and $B \rightarrow C$, we can deduce C .

$$\{A \vee B, A \rightarrow C, B \rightarrow C\} \vdash C$$

- **(for intuitionistic logic) contradiction** If we have $\perp$, then we can deduce any formula A .

$$\perp \vdash A$$

- **(for classical logic) proof by contradiction (implicit proof, or IP)**
If we can deduce $\perp$ from $\neg A$ ($\equiv A \rightarrow \perp$), then we can deduce A .

$$\neg A \dots \perp \vdash A$$

2.6 Practice problems on propositional logic

Show that they are provable in classical propositional logic.

1. $A \vee \neg A$ (**Law of excluded middle**)
2. $(A \vee (B \wedge C) \rightarrow (A \vee C))$
3. $(P \leftrightarrow \neg Q) \rightarrow \neg(P \wedge Q)$
4. $((P \rightarrow Q) \wedge (Q \rightarrow R) \rightarrow (P \rightarrow R))$
5. $(P \rightarrow (Q \wedge R)) \leftrightarrow ((\neg P \wedge Q) \wedge (\neg P \vee R))$

Show that they are provable in minimal propostional logic.

1. $A \wedge B \leftrightarrow B \wedge A$
2. $A \vee B \leftrightarrow B \vee A$
3. $(A \wedge B) \wedge C \leftrightarrow A \wedge (B \wedge C)$ (**Associativity**)
4. $(A \vee B) \vee C \leftrightarrow A \vee (B \vee C)$ (**Associativity**)
5. $A \vee (B \wedge C) \leftrightarrow (A \vee B) \wedge (A \vee C)$ (**Distributivity**)
6. $A \wedge (B \vee C) \leftrightarrow (A \wedge B) \vee (A \wedge C)$ (**Distributivity**)
7. $A \rightarrow (B \rightarrow C) \leftrightarrow (A \wedge B) \rightarrow C$ (**Exportation**)
8. $((A \rightarrow B) \rightarrow (B \rightarrow C)) \rightarrow (A \rightarrow C)$
9. $((A \vee B) \rightarrow C) \leftrightarrow (A \rightarrow C) \wedge (B \rightarrow C)$

10. $\neg(A \wedge \neg A)$
11. $\neg(A \rightarrow B) \rightarrow \neg B$
12. $(A \rightarrow B) \rightarrow (\neg B \rightarrow \neg A)$
13. $A \rightarrow \neg\neg A$
14. $A \vee B \rightarrow \neg(\neg A \wedge \neg B)$
15. $\neg A \vee \neg B \rightarrow \neg(A \wedge B)$
16. $\neg(A \vee B) \leftrightarrow \neg A \wedge \neg B$
17. $\neg A \leftrightarrow \neg\neg\neg A$
18. $(A \rightarrow \neg B) \leftrightarrow (\neg\neg A \rightarrow \neg B)$
19. $\neg\neg(A \vee B) \leftrightarrow \neg(\neg A \wedge \neg B)$
20. $\neg\neg(A \rightarrow B) \rightarrow (\neg\neg A \rightarrow \neg\neg B)$
21. $\neg\neg(A \wedge B) \leftrightarrow (\neg\neg A \wedge \neg\neg B)$

Show that they are provable in intuitionistic propositional logic.

1. $\perp \leftrightarrow A \wedge \neg A$
2. $\neg A \rightarrow (A \rightarrow B)$
3. $(\neg A \vee B) \rightarrow (A \rightarrow B)$
4. $A \vee B \rightarrow (\neg A \rightarrow B)$
5. $\neg(A \rightarrow B) \rightarrow \neg\neg A$
6. $\neg\neg(A \rightarrow B) \leftrightarrow (\neg\neg A \rightarrow \neg\neg B)$
7. $A \vee \perp \leftrightarrow A$
8. $A \wedge \perp \leftrightarrow \perp$

2.7 Semantics on Classical Propositional logic

Now, some of you may have learned how to refute that given propositional formula is not true or not provable. If we give 'truth table' for given formula and we find that there is counterexample, then we just refuted. But there should arise a question : Is any true formula provable in classical logic?

Answer is yes. And it is said that classical logic is complete for truth table semantics. There is another semantics? Yes, and we can find that intuitionistic logic is complete for some different semantics (obviously not for truth table semantics).

First, we prove completeness of classical logic, and in order to do it we need to define some terms.

Definition 2.7. truth table assignment is function v from set of propositional variables to $\mathbb{B} = \{\top, \perp\}$, satisfying following inductive property (actually we need to extend domain of v in a sense, but here we just use same symbol):

1. $v(\top) = \top$
2. $v(\neg A) = \top - v(A)$ (flip)
3. $v(A \wedge B) = \begin{cases} \top, & \text{both } v(A), v(B) = \top \\ \perp, & \text{otherwise.} \end{cases}$
4. $v(A \vee B) = \begin{cases} \top, & \text{both } v(A) = \top \text{ or } v(B) = \top \\ \perp, & \text{otherwise.} \end{cases}$
5. $v(A \rightarrow B) = \begin{cases} \top, & \text{both } v(A) = \perp \text{ or } v(B) = \top \\ \perp, & \text{otherwise.} \end{cases}$

Then we can prove following by using induction on height.

Proposition 2.8. *If two truth assignments v, v' coincide on propositional variables appearing in a formula A , then $v(A) = v'(A)$.*

Here we use $\Gamma \models A$ to denote that any truth assignment that satisfy all formulas in Γ also satisfies A . Then we can prove soundness of truth table semantics of classical propositional logic by using induction again.

Proposition 2.9. *(soundness) For every set Γ of propositional formulas and every propositional formula A , if $\Gamma \vdash A$, then $\Gamma \models A$.*

Now we define some terms to prove the other direction, completeness.

Definition 2.10. We call set of propositional formulas Γ inconsistent, if it proves $\perp$, and consistent otherwise.

Also we call Γ satisfiable if there is a truth assignment that satisfy all formulas in it.

Proposition 2.11. *The following are equivalent in classical, intuitionistic logic.*

1. Γ is inconsistent
2. For every formula A , Γ proves both A and $\neg A$.
3. For every formula A , Γ proves A .

Proposition 2.12. *The following are equivalent.*

1. Γ is unsatisfiable.
2. $\Gamma \models \perp$

Proposition 2.13. *The following are equivalent in classical logic.*

1. $\Gamma \vdash A$
2. $\Gamma \cup \{\neg A\}$ is inconsistent.

Proposition 2.14. *The following are equivalent in classical logic.*

1. $\Gamma \models A$
2. $\Gamma \cup \{\neg A\}$ is unsatisfiable.

These are enough to prove contrapositive of completeness theory. We prove this theorem by introducing maximally satisfiable set of formulas containing Γ .

Proposition 2.15. (completeness) *For every set Γ of propositional formulas and every propositional formula A , if $\Gamma \models A$, then $\Gamma \vdash A$.*

Then there follows compactness theorem.

Proposition 2.16. (Compactness) *For every set Γ of propositional formulas and every propositional formula A , if $\Gamma \models A$, then $\Gamma' \models A$ for some finite subset $\Gamma' \subseteq \Gamma$.*

Example 2.17. Show that propositional formula

$$((A \vee B) \rightarrow C) \rightarrow ((A \rightarrow C) \wedge (C \rightarrow B))$$

is not provable in classical logic, constructing truth assignment that doesn't satisfy it.

2.8 Semantics on intuitionistic Propositional logic

Showing that a formula is not provable in intuitionistic logic is not so easy without constructing new model(semantic) for it. Now i introduce such model, a Kripke model. There is another model Heyting algebra, but i won't deal with it here.

But There is big correspondence : truth table semantic model correspond to Boolelan algebra and filter, Kripke semantic and heyting algebra model correspond to topological interpretation.

Definition 2.18. A Kripke model for intuitionstic logic consists of a partial order $(K, \leq)$ and a function that assigns to each $\alpha \in K$ a truth assignment v_α satisfying the following monotonicity condition: for any proposition P and pair of conditions α, β in K , if $v_\alpha(P) = \top$ and $\beta \leq \alpha$, then $v_\beta(P) = \top$.

We write $\alpha \Vdash P$ and say P is true at α or α forces P when $v_\alpha(P) = \top$. Also we extend truth assignment in similar way as for classical condition:

- $\alpha \Vdash P$ if and only if $v_\alpha(P) = \top$
- $\alpha \Vdash A \wedge B$ if and only if $v_\alpha(A) = \top$ and $v_\alpha(B) = \top$
- $\alpha \Vdash A \vee B$ if and only if $v_\alpha(A) = \top$ or $v_\alpha(B) = \top$
- $\alpha \Vdash A \rightarrow B$ if and only if for every $\beta \leq \alpha$, if $v_\beta(A) = \top$ then $v_\beta(B) = \top$.
- $\alpha \nVdash \perp$.

Here, we can interpret $\beta \leq \alpha$ as β is knowledge acquired in the future further than α . So what inequality says is scale of possibilities. The more knowledge we get, the less possibilies we can think of. Also we call elements of K a node or worlds.

Proposition 2.19. *For any nodes alpha, beta, and any propositional formula A, if $\alpha \Vdash A$ and $\beta \leq \alpha$, then $\beta \Vdash A$.*

We can prove this proposition by induction on formula. Let's consider the case where A was actually conditional $A \rightarrow B$. Suppose $\alpha \Vdash A \rightarrow B$ and $\beta \leq \alpha$. Suppose $\gamma \leq \beta$, and $\gamma \Vdash A$. Since $\leq$ is partial order, it is transitive. Thus we have $\gamma \leq \alpha$. since $\alpha \Vdash A \rightarrow B$, we have $\gamma \Vdash B$.

The reason we modified definition of truth assignment on conditional is due to monotonicity. If we stick to ordinary definition as in classical one, we may

have for $\beta \leq \alpha$, $\alpha \Vdash A \rightarrow B$ But $\beta \nVdash A \rightarrow B$. This happens when $\beta \Vdash A$ But $\alpha \nVdash A$.

How about negation of a formula? $\alpha \Vdash \neg A$ if and only if for every $\beta \leq \alpha$, $\beta \nVdash A$, since $\neg A$ is defined as $A \rightarrow \perp$. So informally speaking, we can say *negA* if in any future, we still find that A is false.

Definition 2.20. We write $\Gamma \models A$ if for every Kripke model $\mathcal{K}$ and node α of $\mathcal{K}$, if $\alpha \Vdash B$ for every $B \in \Gamma$ then $\alpha \Vdash A$.

Now we can prove both soundness and completeness of intuitionistic logic for Kripke semantic.

Proposition 2.21. *If $\Gamma \vdash A$ in intuitionistic or minimal logic, then $\Gamma \models A$ in the corresponding version of Kripke semantics.*

We can use induction on formula.

Now we define saturated set of formulas to make easier proof of completeness. However, it may not be plausible that we can always construct saturated set.

Definition 2.22. A set of formulas Γ is saturated if the following hold :

- Γ is **deductively closed**, in other words if $\Gamma \vdash A$, then $A \in \Gamma$.
- If $A \vee B \in \Gamma$, then $A \in \Gamma$ or $B \in \Gamma$.

Proposition 2.23. *If $\Gamma \nVdash A$, then there is saturated set $\Gamma' \supseteq \Gamma$ such that $\Gamma' \nVdash A$.*

증명 I will show some mathematical technique that is frequently used in mathematical logic, which feels like cheating too. Here is the procedure. First, let Γ_0 be the deductive closure of Γ , that is set of all formulas provable from Γ . Then is Γ_0 deductively closed? Yes. if not, say $\Gamma_0 \vdash A'$, $A' \notin \Gamma_0$, then $\Gamma \vdash A'$ (Check!). So it is contradiction to definition of Γ_0 . Now for second condition of saturated, we consider following pair of disjunctions $(C_i \vee D_i)_{i \in \mathbb{N}}$ of language. Assumption made here is that language is countable. For more cardinality, use zorn's lemma for obtaining saturated sets.

Assuming Γ_i has been made, if $\Gamma_i \nVdash C_i \vee D_i$, set $\Gamma_{i+1} = \Gamma_i$. Otherwise, $\Gamma_i \vdash C_i \vee D_i$, $\Gamma_i \nVdash A$, then there is no possibility for both $\Gamma_i \vdash C_i \rightarrow A$ and $\Gamma_i \vdash D_i \rightarrow A$. So either $\Gamma_i \nVdash C_i \rightarrow A$ or $\Gamma_i \nVdash D_i \rightarrow A$. In the first case, take Γ_{i+1} to be deductive closure of $\Gamma_i \cup C_i$ and vice versa.

Now here trick kicks in. We can't do this procedure for infinitely times but we need to do. There are infinitely many formulas to be discharged or included as done right above. How can we solve this problem? Answer is union on whole natural numbers. It feels cheating but it is mathematically valid.

Take $\Gamma' = \cup_{i \in \mathbb{N}} \Gamma_i$. Then this Γ' is saturated set. $\square$

Now completeness follows directly from above.

Proposition 2.24. (*completeness*)

Let $\mathcal{K}$ be the kripke model whose nodes are all the consistent saturated sets (in other words, it doesn't prove contradiction $\perp$), where $\alpha \leq \beta$ is defined to be $\alpha \supseteq \beta$ and for every variable P , $\alpha \Vdash P$ is defined to hold if and only if $P \in \alpha$. Then we have $\alpha \Vdash A$ if and only if $\alpha \vdash_I A$ for every propositional formula A .

There are many more beautiful, interesting facts about kripke models, check on the textbook if you are interested.

Example 2.25. (Practice Problems) Provide kripke models that show that the following schemas are not intuitionistically valid :

1. $\neg A \vee \neg \neg A$
2. $\neg(A \wedge B) \rightarrow \neg A \vee \neg B$
3. $\neg \neg(A \vee B) \rightarrow \neg \neg A \vee \neg \neg B$
4. $((A \rightarrow B) \rightarrow A) \rightarrow A$
5. $(\neg A \rightarrow B \vee C) \rightarrow (\neg A \rightarrow B) \vee (\neg A \rightarrow C)$
6. $\neg(P \wedge \neg Q) \rightarrow (P \rightarrow Q)$

Proposition 2.26. (*proof of Gilvenko's theorem*) Show that a propositional formula A is provable classically if and only if $\neg \neg A$ is provable in intuitionistic logic.

증명 First, Assume $\not\vdash_I \neg \neg A$ (Intuitionistically not provable). Then by completeness of intuitionistic logic, for some model $\mathcal{K}$, for all $\alpha \in \mathcal{K}$, $\alpha \not\vdash \neg \neg A$, there is a node $\beta \in \mathcal{K}$ such that $\beta \Vdash \neg A$.

Now, consider minimal node α , that is, for all $\beta \leq \alpha$, $\beta = \alpha$.

We can show that $\alpha \Vdash A$ if and only if $v_\alpha \models A$ on induction on formula.

We can also show that propositional formula is provable in intuitionistic logic if and only if it is true in every finite kripke model. So take finite model of $\mathcal{K}$, so that we can find a minimal node α .

So α , we have $\alpha \Vdash \neg A$ so we have $\alpha \not\models A$ (check!), so it classically has counterexample. So not derivable in classical logic due to completeness of classical logic.

□

2.9 Practice problem on semantics of propositional logic

Show that following formula is not provable in classical logic by counter example :

1. $((P \vee Q) \rightarrow R) \rightarrow (P \wedge (Q \rightarrow R))$
2. $((P \rightarrow Q) \vee (P \rightarrow R)) \rightarrow (P \rightarrow (Q \wedge R))$
3. $((P \rightarrow Q) \wedge (Q \rightarrow R) \wedge (R \rightarrow P)) \rightarrow ((P \wedge Q) \wedge R)$ (interpretation of hangman paradox?)
4. $((P \leftrightarrow Q) \rightarrow R) \wedge ((Q \leftrightarrow R) \rightarrow P) \rightarrow ((P \wedge Q) \wedge R)$
5. $((P \rightarrow Q) \rightarrow (Q \rightarrow P)) \rightarrow (P \leftrightarrow Q)$
6. $(P \rightarrow (Q \rightarrow R)) \rightarrow ((P \vee Q) \rightarrow R)$
7. $((P \rightarrow Q) \wedge (P \rightarrow \neg Q)) \rightarrow Q$

Show that whether following provability relation is true in classical logic and if not, construct counter example :

1. $\{P \vee Q, R \wedge Q, K \rightarrow R\} \vdash \neg K \rightarrow P$
2. $\{R \vee (P \wedge Q), K \leftrightarrow S, R \wedge S\} \vdash K \rightarrow (P \vee Q)$
3. $\{I \vee (J \wedge K), (\neg I \rightarrow J) \rightarrow L, L \leftrightarrow (\neg J \vee \neg I)\} \vdash (I \vee K) \wedge (I \rightarrow \neg J)$
4. $\{B \wedge (D \vee C), D \rightarrow (A \vee E), \neg E \vee (B \wedge C)\} \vdash (A \rightarrow E) \vee C$

3 First order logic

3.1 Deduction on first order logic

With deduction rules of propositional logic we just learned, we can now extend them into that of first order logic, regarding propositional formula as that in first order logic, but there are problems with variables and quantifier. So how do we deal with this quantifier problem? we need to classify *type* of a variable.

For example, in formula $(f(x) = c)$, we call x a *free variable* of the formula, and if formula is of form $\exists x(f(x) = c)$, then we call x *bound* or not free variable. In this sense, if x is bound variable, we can substitute x to s or t or whatever not used in the formula, since it is irrelevant with the fact the *something* exists that satisfy the formula. and we note that if there is any bound part of a variable x in formula A , then x is not free in A .

Definition 3.1. Γ is set of all constants, function symbols, Δ is set of all relational(predicate) symbols.

A term is defined inductively as :

- Each variable x is a term.
- If $c \in \Gamma$ is a constant symbol, then c is a term.
- If $f \in \Gamma$ is a k -ary function symbol with $k > 0$, and $t_0, \dots, t_{k-1}$ are all terms, then so is $f(t_0, \dots, t_{k-1})$.

Definition 3.2. (A well formed formula) Formula is defined inductively as :

- $\perp$ is a formula
- If s, t are terms then $s = t$ is a (**atomic**) formula.
- If $t_0, \dots, t_{k-1}$ are terms and R is a k -ary relation(predicate) symbol of L , then $R(t_0, \dots, t_{k-1})$ is a (**atomic**) formula.
- If A, B are formula, then so are $(A \wedge B), (A \vee B), (A \rightarrow B)$.
- If A is a formula and x is a variable, then $(\exists x A)$ and $(\forall x A)$ are formulas.

Now we will formally introduce this rule in dedection rule of first order logic. A **free** variable of formula A is a variable, which is bounded by quantifier.

Definition 3.3. A set of free variable $F(A)$ for a given formula A , is defined inductively :

- $F(\perp) = \emptyset$
- $F(s = t) = F(s) \cup F(t)$
- $F(R(t_0, \dots, t_{k-1})) = F(t_0) \cup \dots \cup F(t_{k-1})$
- $F(A \vee B) = F(A \wedge B) = F(A \rightarrow B) = F(A) \cup F(B)$
- $F(\forall x A) = F(\exists x A) = F(A) - \{x\}$

Example 3.4. Find all free variables in given formula :

1. $\exists y(f(x, y) = f(f(y), x))$
2. $\forall x(R(x, y)) \wedge \exists y(f(y) = x)$
3. $\forall x[f(x) = y \vee \forall y(R(y) \rightarrow (f(x) \neq y))]$

We will write $A[t/x]$ to denote we substitute every x in the formula to t .

Now we give rules(Here, t is term without any variable.):

- $\forall x A \rightarrow A[t/x]$
- $A[t/x] \rightarrow \exists x A$.
- If x is not free in A , from $A \rightarrow B$ conclude $A \rightarrow \forall x B$.
- If x is not free in B , from $A \rightarrow B$ conclude $\exists x A \rightarrow B$.
- If x is not free in A , y is not free in B then from $A \rightarrow B$ conclude $A \rightarrow \forall y B[y/x]$.
- If x is not free in B and y is not free in A , then from $A \rightarrow B$ conclude $\exists y A[y/x] \rightarrow B$.

We need to very careful about free variable conditions. Consider following example :

$$x = 2 \rightarrow x = 2, \text{ conclude } x = 2 \rightarrow \forall x, x = 2$$

Then we can interpret (we will define interpretation later,) formula as if quantifier is ranging over natural numbers and 2 is interpreted as ordinary number 2. Then this is absurd. If something is 2, then every natural number is 2? If x is not free, then this means we are justified enough to generalize conclusion, in other words we can put universal quantifier on formula B .

Also we need to be careful about conditional proof. For example, for every natural number bigger than 1, there is a natural number bigger than 1 that can divide that number. We can write this as using relation symbol $R(x, y)$ as $x > 1$ and $x|y$ then $\forall x \exists y R(x, y)$ is true in natural number. But that doesn't mean we can conclude $\exists x \forall y R(x, y)$. If you instantiate x, y and generalize y , this causes this problem.

Now we add following rules with assumption that x is not free in B or any hypothesis other than A .

$$\frac{A}{\forall x A} \forall I \quad \frac{\forall x A}{A[t/x]} \forall E \quad \frac{A[t/x]}{\exists x A} \exists E \quad \frac{\overline{A} \quad \vdots \quad \exists x A \quad B}{B} \exists E$$

Now for elimination rule, we note in sequent form (**Careful! notice the difference between $A[t/x]$ and $A(x)$. Since t is specific term, you can not generalize it. in other words, $A[t/x] \rightarrow \forall x A$ is prohibited. Also, when we generalize variable or a term, there should be no free variable in hypothesis $\Gamma...$ Also, term in elimination rule of $\exists$ is restricted that it must not have any variable):**

$$\frac{\Gamma \vdash A}{\Gamma \vdash \forall x A} \forall R \quad \frac{\Gamma \vdash \forall x A}{\Gamma \vdash A[t/x]} \forall E$$

$$\frac{\Gamma \vdash A[t/x]}{\Gamma \vdash \exists x A} \exists I \quad \frac{\Gamma \vdash \exists x A \quad \Gamma, A \vdash B}{\Gamma \vdash A[t/x]} \exists E$$

Up to now, we didn't consider rule for equality. Intuitively, equal terms must have same property. If $s = t$, then if $A(s)$ is derivable, then $A(t)$ must be derivable too. (Here $A(x)$ denotes that formula A has x as free variable.)

2. First Order Logic

Standard Natural Deduction Form

$$\begin{aligned} \text{Universal Intro } (\forall I) : & \quad \frac{A}{\forall x A} \quad (\text{단, } x \text{는 가정에서 자유 변수가 아님}) \\ \text{Universal Elim } (\forall E) : & \quad \frac{\forall x A}{A[t/x]} \forall E \\ \text{Existential Intro } (\exists I) : & \quad \frac{A[t/x]}{\exists x A} \exists I \\ \text{Existential Elim } (\exists E) : & \quad \frac{\exists x A \quad [A] \cdots B}{B} \exists E \quad (\text{단, } x \text{는 } B \text{에서 자유 변수가 아님}) \end{aligned}$$

Sequent Form ($\Gamma \vdash A$)

$$\begin{aligned} \forall R \text{ (Right)} : & \quad \frac{\Gamma \vdash A}{\Gamma \vdash \forall x A} \\ \forall E \text{ (Elim)} : & \quad \frac{\Gamma \vdash \forall x A}{\Gamma \vdash A[t/x]} \\ \exists I \text{ (Intro)} : & \quad \frac{\Gamma \vdash A[t/x]}{\Gamma \vdash \exists x A} \\ \exists E \text{ (Elim)} : & \quad \frac{\Gamma \vdash \exists x A}{\Gamma \vdash A[t/x]} \end{aligned}$$

Definition 3.5. Equality axioms are universal closure of the following axioms :

- $x = x$
- $x = y \rightarrow y = x$
- $x = y \wedge y = z \rightarrow x = z$
- $x_0 = y_0 \wedge x_1 = y_1 \wedge \dots \wedge x_{k-1} = y_{k-1} \rightarrow f(x_0, \dots, x_{k-1}) = f(y_0, \dots, y_{k-1})$
for each function symbol f
- $x_0 = y_0 \wedge x_1 = y_1 \wedge \dots \wedge x_{k-1} = y_{k-1} \rightarrow R(x_0, \dots, x_{k-1}) = R(y_0, \dots, y_{k-1})$
for each relation symbol R

Proposition 3.6. *The following are provable in minimal logic for any term $t(x)$ and formula $A(x)$:*

1. $x = y \rightarrow t(x) = t(y)$
2. $x = y \rightarrow (A(x) \rightarrow A(y))$

증명 We can prove using induction on term and formula. Recall how we constructed term and formulas inductively. □

3.2 Practice problem on first order logic

Prove the following provability relation on classical logic (Here $(,)$ are omitting, a, b, c, d are constant, others are variables).

1. $\{\forall x(\exists yRxy \rightarrow \exists zRzx), \exists x(Rxb \wedge x = d)\} \vdash \exists zPzd$
2. $\{\exists(Px \wedge Rx), \forall x(Px \rightarrow \forall y(Qy \rightarrow x = y)), Qa\} \vdash Pa \wedge Ra$
3. $\{\forall x(Rx \vee (Px \wedge Qx)), \neg(Ra \vee Pb)\} \vdash \neg a = b$
4. $\{\forall x((Px \vee Qx) \rightarrow Rx), \neg Rc, Pd\} \vdash \neg c = d$
5. $\{\forall x(Px \vee (Qx \wedge Rx)), \neg(\forall x)Qx, \forall x(Px \rightarrow x = c)\} \vdash \exists xx = c$
6. $\{\forall x((Px \wedge Qx \wedge \neg x = a) \rightarrow \exists yRxy), \neg \exists yRby, Sa \wedge \neg Sb\} \vdash \neg(Pb \wedge Qb)$
7. $\{Pa \wedge \exists x(Qx \wedge \neg Rxa), \forall x(Sx \rightarrow \forall yRyx)\} \vdash \exists x(Px \wedge \neg Sx)$
8. $\{\forall x(Px \rightarrow Pf(x)), Pa, f(a) = b\} \vdash Pb$

Show that the following logical truth in classical logic (All symbols are either variables or relation(predicate) symbol.):

1. $\forall x \forall y R(x, y) \rightarrow \exists x \exists y R(x, y)$
2. $\forall x \exists y (R(x, y) \vee \neg R(x, y))$
3. $\forall x (P(x) \rightarrow (\forall y Rxy)) \leftrightarrow \forall x \forall y (P(y) \rightarrow R(y, x))$

Show that the following are derivable in minimal logic :

1. $\neg \exists x \forall y (\neg S(y, y) \leftrightarrow S(x, y))$ (**Barber's paradox**)
2. $\forall x A \leftrightarrow A$ if x is not free in A
3. $\exists x A \leftrightarrow A$ if x is not free in A
4. $\forall x (A \wedge B) \leftrightarrow \forall x A \wedge \forall x B$
5. $\exists x (A \wedge B) \leftrightarrow (\exists x A) \wedge B$ if x is not free in B
6. $\forall x A \vee B \rightarrow \forall x (A \vee B)$ if x is not free in B
7. $\exists x (A \vee B) \leftrightarrow \exists x A \vee \exists x B$
8. $\forall x (A \rightarrow B) \leftrightarrow (\exists x A \rightarrow B)$ if x is not free in B
9. $\exists x (A \rightarrow B) \rightarrow (\forall x A \rightarrow B)$ if x is not free in B (**Drinker's paradox**)
10. $\forall x (A \rightarrow B) \leftrightarrow (A \rightarrow \forall x B)$ if x is not free in A
11. $\exists x (A \rightarrow B) \rightarrow (A \rightarrow \exists x B)$ if x is not free in B
12. $\neg \exists x A \leftrightarrow \forall x \neg A$
13. $\exists x \neg A \rightarrow \neg \forall x A$

Show that the following are derivable in classical logic :

1. $\exists x A \leftrightarrow \neg \forall x \neg A$
2. $\forall x A \leftrightarrow \neg \exists x \neg A$
3. $\forall x (A \vee B) \leftrightarrow \forall x A \vee B$ if x is not free in B
4. $\exists x (A \rightarrow B) \leftrightarrow (\forall x A \rightarrow B)$ if x is not free in B
5. $\exists x (A \rightarrow B) \leftrightarrow (A \rightarrow \exists x B)$ if x is not free in A

Prove Double negation shift theorem :

Let DNE be the schema $\forall x \neg \neg A(x) \rightarrow \neg \neg \forall x A(x)$.

Then show that Classical first order logic proves a formula B if and only if Intuitionistic + DNE proves $\neg \neg B$.

3.3 Semantics of classical first order logic

In First order language, we often use semantic L -structure $\mathfrak{M}$, known as Tarski's model semantic. We give interpretation to formula for each and every symbol except for variable. Then we investigate truth value of that interpretation.

For example, $\exists x(f(x) = 1)$ can be interpreted as :

- f is natural function $f : \mathbb{N} \rightarrow \mathbb{N}, f(n) = n^2$.
- 1 is interpreted as ordinary natural number 1.
- quantifier ranger over $\mathbb{N}$. In other words, x is element of set of natural number.
- Then this interpretation gives true value to the formula, since $1^2 = 1$.

Thus, we need to set something before we give full interpretation. First, we need to specify our domain (or universe of a structure, usually **we use** $|\mathfrak{M}|$ for **universe** of $\mathfrak{M}$). Second, we must specify how we interprete all the functio, relation, constant symbols in the language. And finally, we need to give assignments to all free variable from the universe. But we treat differntly about conditional or quantified formula.

Here we use $\sigma : \text{variables} \rightarrow |\mathfrak{M}|$ to denote assignment function for free variables. then $\mathfrak{M} \models_{\sigma} A$ means from L structure $\mathfrak{M}$, one specific assignment satisfies formula A . You can imagine assignment as plugging in a real value to variable symbol.

Definition 3.7. Following are only way to give truth values to given formula from L -structure $\mathfrak{M}$:

- $\mathfrak{M} \models_{\sigma} t = t'$ if and only if $[t]^{\sigma} = [t']^{\sigma}$.
- $\mathfrak{M} \models_{\sigma} R(t_0, \dots, t_{k-1})$ if and only if $R^{\mathfrak{M}}([t_0]^{\sigma}, \dots, [t_{k-1}]^{\sigma})$
- $\mathfrak{M} \models_{\sigma} \perp$ is always false.
- $\mathfrak{M} \models_{\sigma} A \wedge B$ if and only if $\mathfrak{M} \models A$ and $\mathfrak{M} \models B$.
- $\mathfrak{M} \models_{\sigma} A \vee B$ if and only if $\mathfrak{M} \models A$ or $\mathfrak{M} \models B$.
- $\mathfrak{M} \models_{\sigma} A \rightarrow B$ if and only if $\mathfrak{M} \not\models A$ or $\mathfrak{M} \models B$.
- $\mathfrak{M} \models_{\sigma} \exists x A(x)$ if and only if there is $a \in |\mathfrak{M}|$ such that $\mathfrak{M} \models_{\sigma[x \mapsto a]} A$.
- $\mathfrak{M} \models_{\sigma} \forall x A(x)$ if and only if for all $a \in |\mathfrak{M}|$, $\mathfrak{M} \models_{\sigma[x \mapsto a]} A$.

Find counterexample of the inference(Language is open. Just think of it as minimal as possible. Also all the symbols are either constant, variable, or relation symbols):

1. $\forall x \exists y R(x, y) \rightarrow \exists x \forall y R(x, y)$
2. $\Gamma = \{\forall x(P(x) \rightarrow \exists y R(x, y)), \exists x R(a, x)\} \vdash \exists \neg P(x)$
3. $\Gamma = \{\forall x(P(x) \rightarrow Q(x, c)), \exists x(Q(c, x) \wedge P(x)), P(a)\} \vdash Q(c, a)$
4. $\Gamma = \{\exists x \exists y(D(x, y) \wedge \neg D(x, y)), D(a, b)\} \vdash \neg D(b, a)$
5. $\Gamma = \{(\exists x)[\forall y(D(y) \rightarrow F(y, x)) \vee \forall y(E(y) \rightarrow F(y, x))], \forall x(Dx \vee Ex)\} \vdash \exists x F(x, x)$

3.4 Semantics of intuitionistic first order logic

As in propositional logic case, we use kripke semantics here. Only thing we need to add is rules about equality and quantifier.

Definition 3.8. A kripke structure $\mathfrak{K}$ for a language L consists of partial order $(K, \leq)$ together with assignment of a structure(Tarski), for every node $\alpha \in K$, $\mathfrak{K}_\alpha$, in such a way that it holds for every $\beta \leq \alpha$, function symbol f and relation(predicate) symbol R ,

- $|\mathfrak{K}_\beta| \supseteq |\mathfrak{K}_\alpha|$
- If $a_0, \dots, a_{k-1} \in |\mathfrak{K}_\alpha|$, then $f^{\mathfrak{K}_\alpha}(a_0, \dots, a_{k-1}) = f^{\mathfrak{K}_\beta}(a_0, \dots, a_{k-1})$
- If $a_0, \dots, a_{k-1} \in |\mathfrak{K}_\alpha|$, and if $R^{\mathfrak{K}_\alpha}(a_0, \dots, a_{k-1}) = R^{\mathfrak{K}_\beta}(a_0, \dots, a_{k-1})$

Similarly, we give satisfying rule for kripke model :

Definition 3.9. We define forcing relation $\Vdash$ using recursion on formulas.

- $\alpha \Vdash_\sigma A$ if and only if $\mathfrak{K}_\alpha \models_\sigma A$, when A is atomic formula.
- $\alpha \Vdash_\sigma A \wedge B$ if and only if $\alpha \Vdash A$ and $\alpha \Vdash B$
- $\alpha \Vdash_\sigma A \vee B$ if and only if $\alpha \Vdash A$ or $\alpha \Vdash B$
- $\alpha \Vdash_\sigma A \rightarrow B$ if and only if $\forall \beta \leq \alpha, (\beta \Vdash_\sigma A \rightarrow \beta \Vdash_\sigma B)$.
- $\alpha \Vdash_\sigma \exists x A(x)$ if and only if $\exists a \in |\mathfrak{K}_\alpha|, (\alpha \Vdash_{\sigma[x \mapsto a]} A)$
- $\alpha \Vdash_\sigma \forall x A(x)$ if and only if $\forall \beta \leq \alpha, \forall a \in |\mathfrak{K}_\beta|, (\beta \Vdash_{\sigma[x \mapsto a]} A)$

- $\alpha \not\Vdash_{\sigma} \perp$

Then we can do similar things as we do in propositional logic.

Proposition 3.10. *For every $\beta, \alpha, \beta \leq \alpha$ and σ , if $\alpha \Vdash_{\sigma} A$ then $\beta \Vdash_{\sigma} A$.*

증명 We can use induction on formulas and it is easy to check validity for quantifier case. Other cases are already done in propositional logic. $\square$

Now we prove soundness for kripke semantics on intuitionistic logic. We can use induction and it is easy to check. Here $\Gamma \Vdash A$ is defined similarly as in propositional case : $\alpha \Vdash \Gamma \Rightarrow \alpha \Vdash A$, for all α

Proposition 3.11. *(Soundness)*

For every set of sentences Γ and every sentence A , if $\Gamma \vdash A$, then $\Gamma \Vdash A$.

Show that followings are not derivable in intuitionistic logic by constructing kripke model counterexample.

1. $\forall x(A \vee B(x)) \rightarrow A \vee \forall x B(x)$, where x is not free in A
2. $(A \rightarrow \exists x B(x)) \rightarrow \exists x(A \rightarrow B(x))$
3. $\neg \neg \forall x(B(x) \vee \neg B(x))$
4. $\neg \forall x A(x) \rightarrow \exists x \neg A(x)$
5. $(A \rightarrow \exists x B(x)) \rightarrow \exists x(A \rightarrow B(x))$
6. $\forall x \neg \neg A(x) \rightarrow \neg \neg \forall x A(x)$.

3.5 Completeness and compactness theorem of first order logic

Again, we prove completeness, and hence compactness in first order classical, intuitionistic logic, respectively.

Theorem 3.12. *(Completeness for classical first order logic) For every set of sentences Γ , a sentence A , if $\Gamma \models A$ then $\Gamma \vdash A$ in classical logic.*

If we can prove this, we have Compactness.

Theorem 3.13. *(Compactness theorem) If every finite subset of Γ has a model, then Γ has a model.*

증명 Suppose it doesn't have model. Then we have $\Gamma \models \perp$, thus $\Gamma \vdash \perp$ by completeness. Then since proof is finite, we have $\Gamma_0 \vdash \perp$ for some finite subset $\Gamma_0 \subseteq \Gamma$. So for this Γ_0 , we have $\Gamma_0 \models \perp$ by soundness, thus doesn't have model. We have just proved contrapositive of the statement. $\square$

증명 (Proof of completeness theorem) Proof is quite hard and complicated, and even raises question. As done in propositional logic case, we prove contrapositive version :

$$[\Gamma \not\models A \Rightarrow \Gamma \not\vdash A] \iff [\Gamma \cup \{\neg A\} \text{ is consistent} \Rightarrow \Gamma \cup \{\neg A\} \text{ has a model}]$$

Now we assume that $\Gamma \cup \{\neg A\}$ is consistent. Now we must construct a model of $\Gamma \cup \{\neg A\}$.

As you may have experienced, a model can be anything. finite, infinite, real numbers, natural numbers.. even logic system itself. So what we do here is to construct model, based on our logic itself. Goal is to construct L structure $\mathfrak{M}$ satisfying following :

$$\mathfrak{M} \models A \iff A \in \Delta$$

For some set of formula Δ . So whole completeness of first order logic is based on set theory(by usage of symbol $\in$) afterall. and set theory is based on first order logic... can you see the point?

Anyway, how do we construct such $\mathfrak{M}$ and Δ ? The procedure is called Henkin construction. It again uses union on natural numbers and some weird one, Zorn's lemma. Zorn's lemma is used many time in mathematics and logic. If you are interested in the completeness proof (Henkin construction), read Avigad's or Martin Hils' textbook A first journey through logic.

Outline of proof : We will construct model from logic language itself. So whole process is quite similar to that of propositional logic. But we need to add one more thing, rules for quantifier. We call T a **henkin theory**, if for any $A(x)$, there is constant symbol c such that $\exists x A(x) \rightarrow A(c) \in T$. Then first we need to build henkin theory from the given base theory, and we construct complete henkin theory to use method mentioned above. $\square$

4 Axiomatic Set theory

4.1 What is axiom?

Axiom is a rule that explains (not straightforwardly, rather concisely) given field. For example for those who know axioms of group, we can express it as a first-order language :

1. $\forall xyz, (f(f(x, y), z) = f(x, f(y, z)))$
2. $\forall x, (f(x, e) = f(e, x) \wedge f(x, e) = x)$
3. $\forall x, \exists y(f(x, y) = e)$

We can also write Peano axioms in first order language by adding some constant 0 and function $S : \mathbb{N} \rightarrow \mathbb{N}$.

Axioms are not straightforward. Their appearance isn't intuitive and doesn't give us much of idea. But after learning some theories on those axioms, we can understand them and use it freely later on.

Now we state axioms of set theory in first order language, and prove some interesting theories.

4.2 Axioms and language of set theory

Here i will first enumerate all the axioms of Zermelo-frankel set theory.

- 1.

4.3 Set and Proper class

4.4 How to formalize mathematics